ELT3207 Meas. & Digital Control Basics

Lecture 9: Design of digital control systems

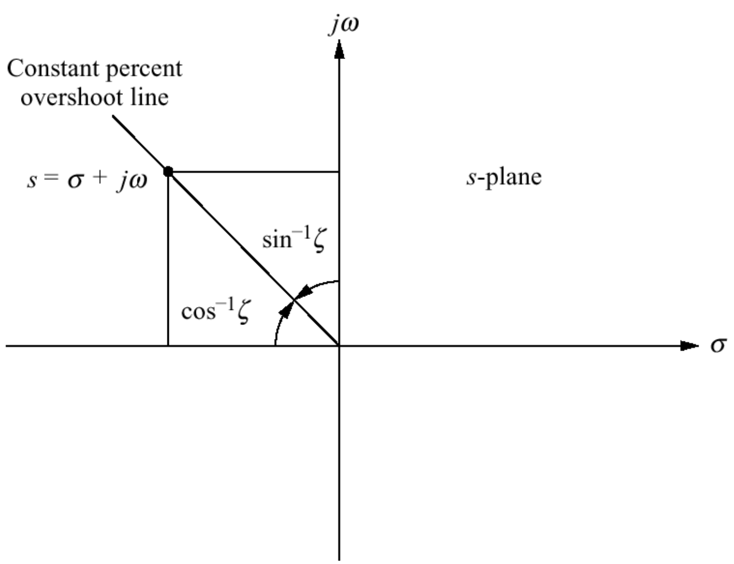
Hoang Gia Hung
Faculty of Electronics and Telecommunications
University of Engineering and Technology, VNU Hanoi

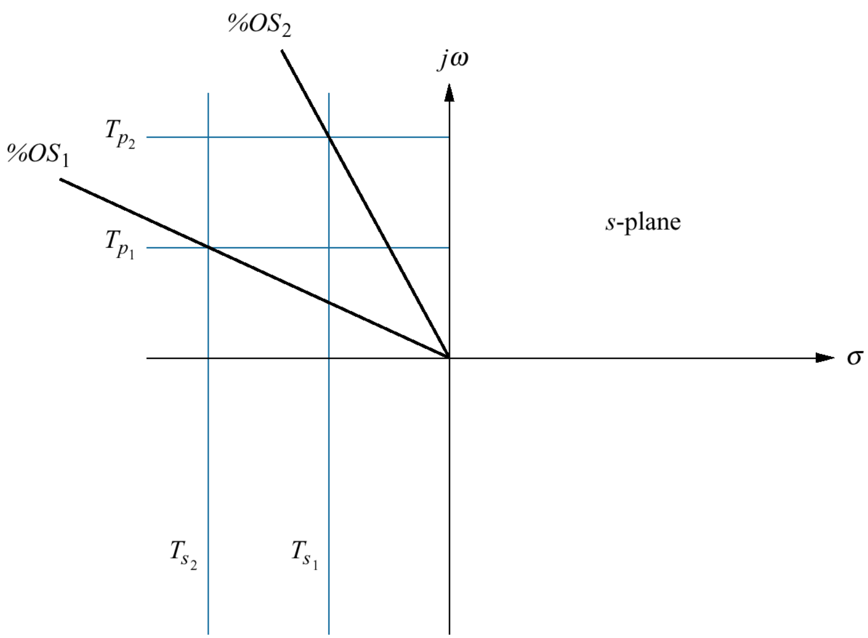
Last lecture review

- **Advantages of digital control systems:**
  - Can control numerous loops at reduced cost.
  - System modifications can be implemented with software changes.
- **The z-transform.**
  - Allows us to represent sampled waveforms at the sampling instants
  - Allow handling of sampled systems as easily as continuous systems, including block diagram reduction
- **Stability analysis**
  - Can be done in the z-pane: the unit circle becomes the boundary of stability
  - Can also be done in the s-plane via bilinear transformation
- **Steady-state error analysis**
  - Depends on the placement of the sampler, but parallels the s-plane techniques.
- **Today's lecture**: Digital controller synthesis.

Transient response for analog systems (recap)

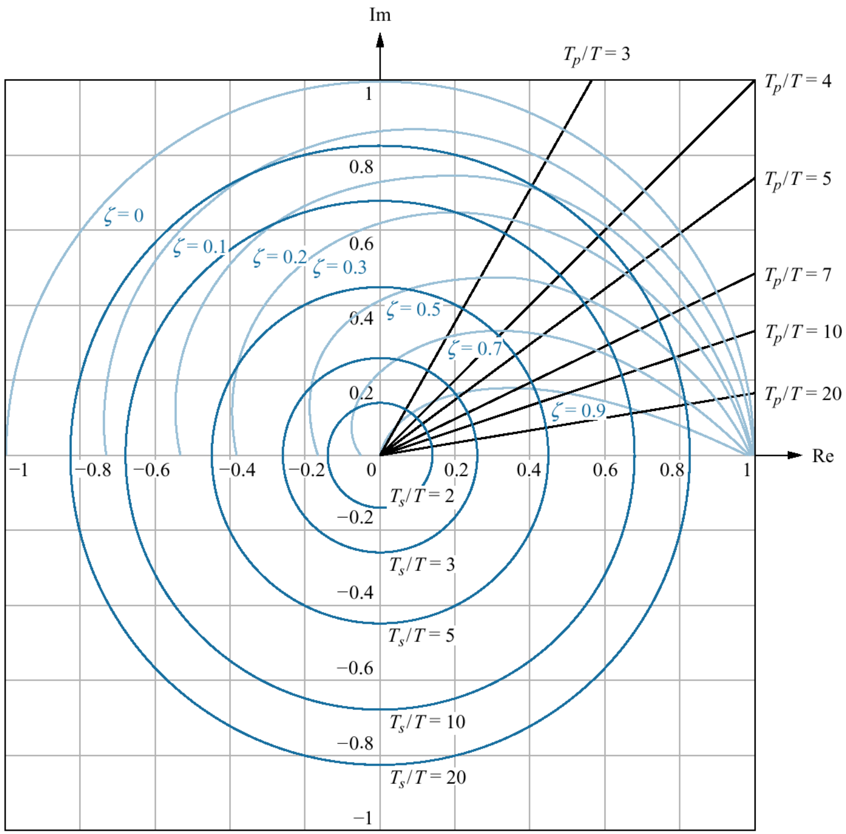
- Relationship between transient response & s-plane (ELT3051)
  - vertical lines on the s-plane were lines of constant settling time
  - horizontal lines were lines of constant peak time
  - radial lines were lines of constant percent overshoot
- How do we establish equivalent relationship on the z-plane?





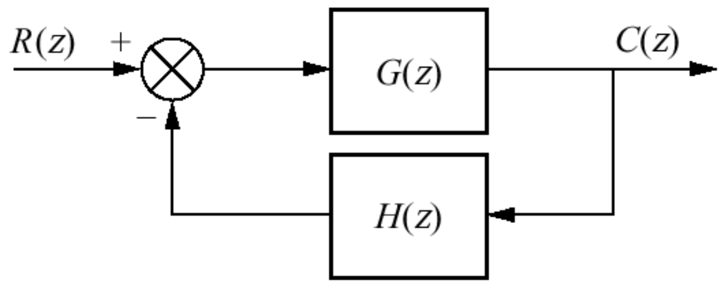
Transient response for digital systems

- Mapping lines on the s-plane to the z-plane through $z = e^{sT}$
  - E.g., lines of constant settling time: $s = \sigma_1 + j\omega \rightarrow z = e^{\sigma_1 T} e^{j\omega T} = r_1 e^{j\omega T} \rightarrow$ concentric circles of radius $e^{\sigma_1 T} = e^{-4/(T_s/T)}$. Also $\frac{T_s}{T} = -4/\ln(r)$.



Constant damping ratio, normalized settling time, and normalized peak time plots on the *z*-plane.

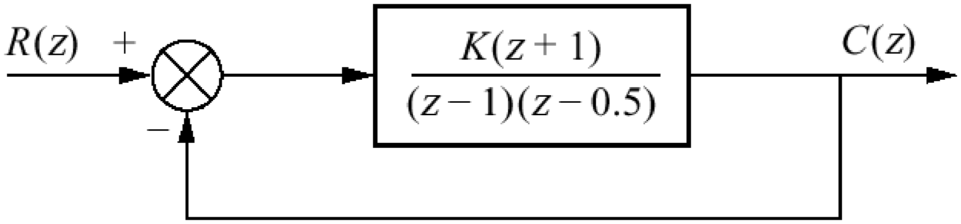
Gain design on the z-plane



- Open-loop and closed-loop transfer functions for digital systems are identical to continuous systems, except variable change $s \to z$
  - Can use the same rules for plotting a root locus and determine the gain required for stability/transient response requirements.
- **Stability design**: region of stability on the z-plane is within the unit circle and not the left half-plane.
  - To determine stability, we must search for the intersection of the root locus with the unit circle rather than the imaginary axis.
- **Transient response design**: find the intersection of the root locus with the appropriate curves ($T_s/T_p/\zeta$) on the z- plane.

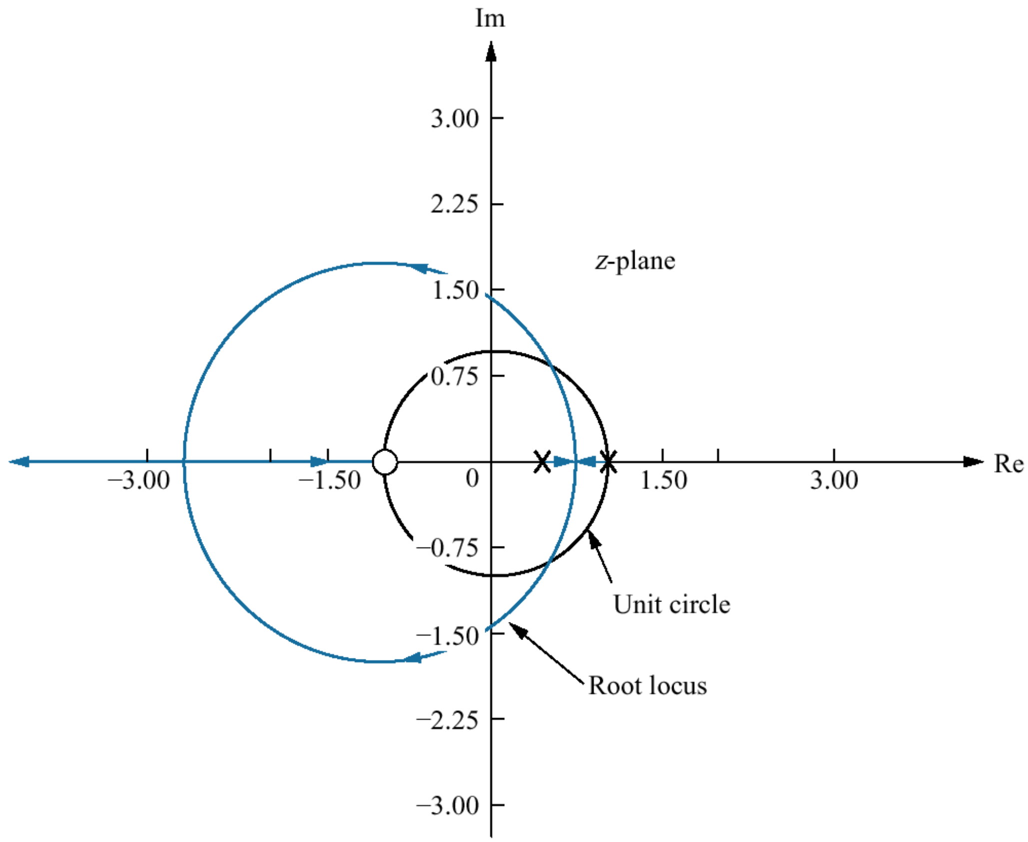
Gain design: example

- Sketch the root locus for the below system and
  a. Determine the range of gain, K, for stability from the root locus plot.
  b. Find the value of gain to yield a damping ratio of 0.7.



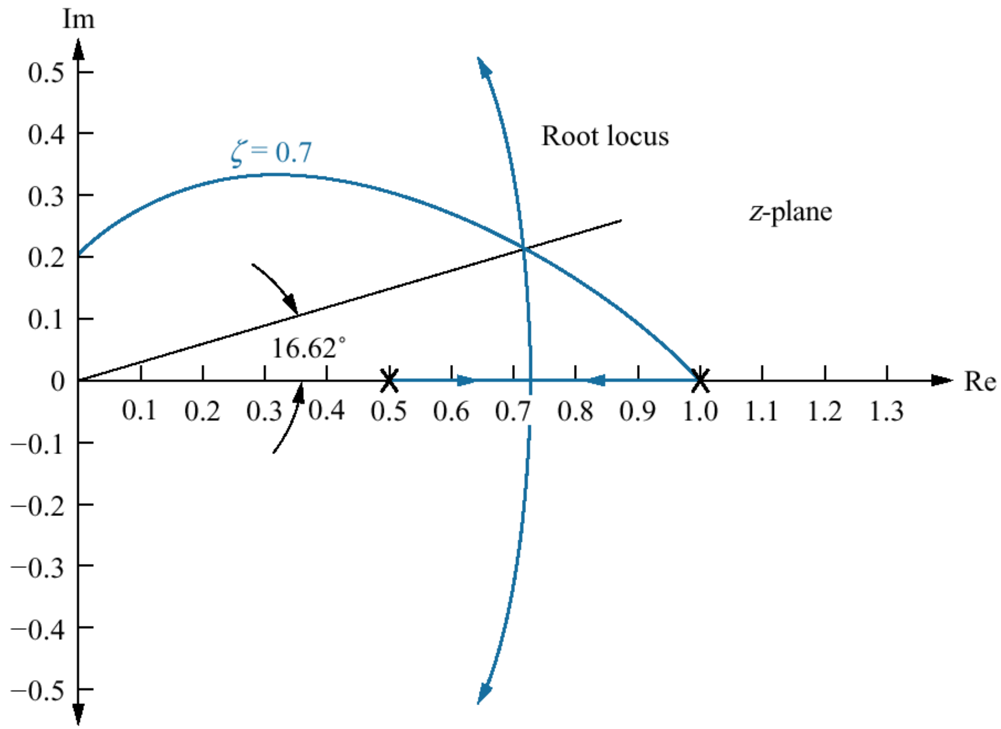
- **Solution**:
  - Treat the system as if z were s, and sketch the root locus.

Example 1: solution (a)



- Search along the unit circle for a total angle of the zeros & poles of $-180°$ → the intersection with the root locus is $1\angle 60°$.
- The gain at intersection point is 0.5 → the range of gain for stability is $0 < K < 0.5$

Example 1: solution (b)



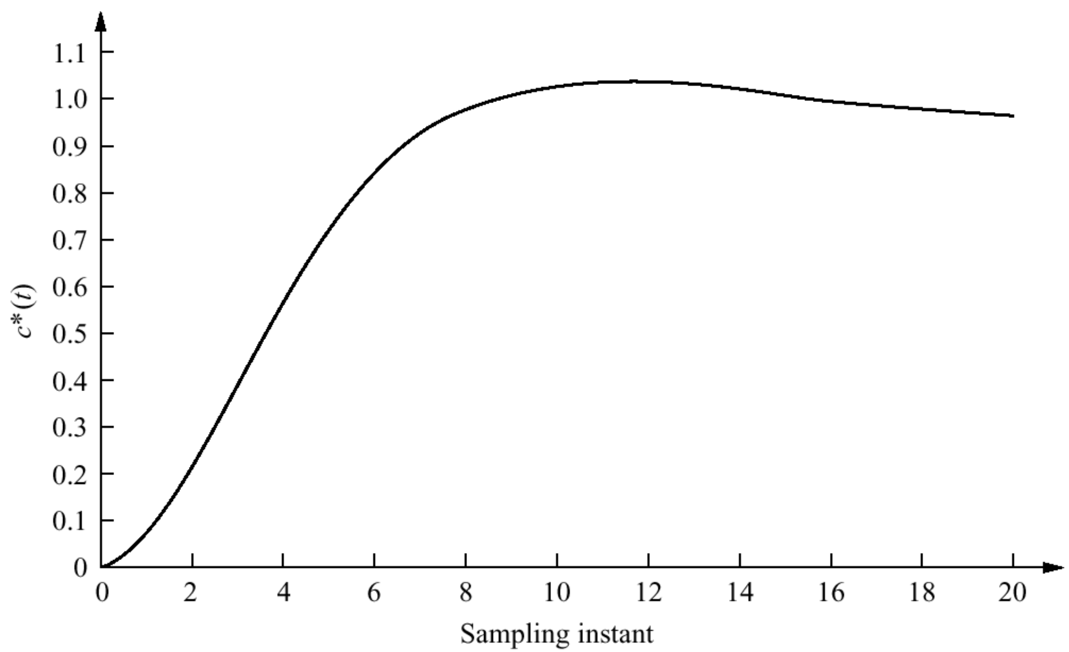
- Draw a radial line from the origin to the intersection of the root locus with the 0.7 damping ratio curve → the angle of the line is $16.62°$.
- Search along the $16.62°$ line for a total angle of the zeros & poles of $-180°$ → the intersection is at $0.719 + j0.215$ → the gain at intersection point is $K = 0.0627$.

Example 1: solution (b)

- **Verification**: for $K = 0.0627$, the unit sampled step response of the system is

$$C(z) = \frac{R(z)G(z)}{1+G(z)} = \frac{\left[\frac{z}{z-1}\right]\left[\frac{0.0627(z+1)}{(z-1)(z-0.5)}\right]}{1+\frac{0.0627(z+1)}{(z-1)(z-0.5)}} = \frac{0.0627z^2+0.0627z}{z^3-2.4373z^2+2z-0.5627}.$$

- Since $\%OS \approx 5\%$, the requirement of a 0.7 damping ratio has been met.

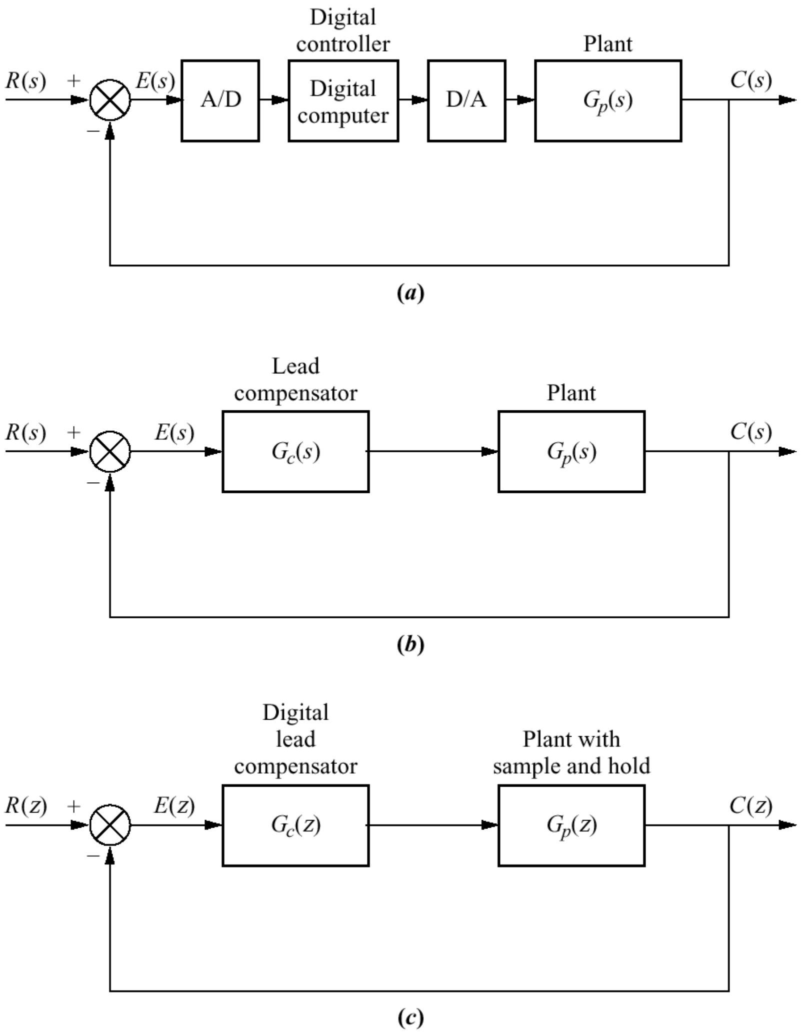


Note: Valid only at integer values of sampling instant.

Cascade compensation (1)

a. Digital control system showing the digital computer performing compensation;
b. continuous system used for design;
c. transformed digital system via an appropriate bilinear transformation.

- The bilinear transformation covered last week will <u>**not**</u> preserve the response of the continuous compensator.
- We will use **Tustin transformation** instead: $s = \frac{2(z-1)}{T(z+1)}$ & $z = \frac{-\left(s+\frac{2}{T}\right)}{s-\frac{2}{T}} = \frac{1+\frac{T}{2}s}{1-\frac{T}{2}s}$

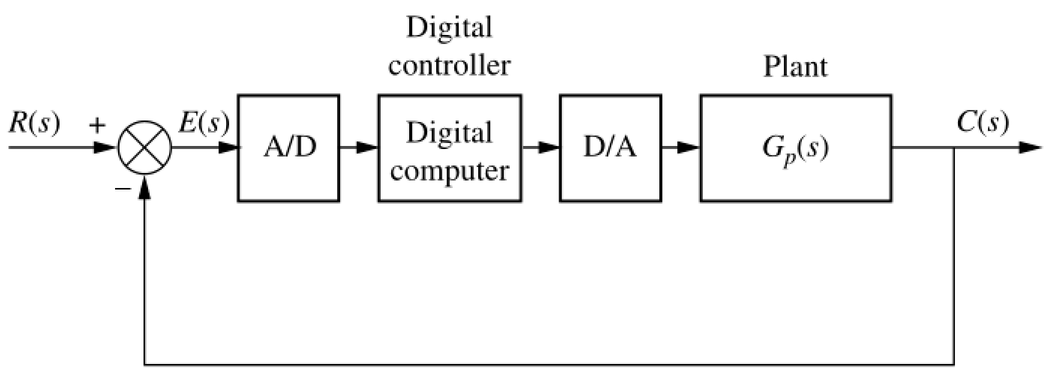


Cascade compensation (2)

- How do we choose the sampling interval when designing digital compensators?
  - Smaller sampling interval (higher sampling rate) → closer match to the analog compensator.
  - Lower sampling rate → discrepancy at higher frequencies between the digital and analog filters' frequency responses.
  - **General rule**: the lowest value of T in seconds should be in the range $0.15/\omega_{\Phi_M}$ to $0.15/\omega_{\Phi_M}$, where ω_{Φ_M} is the zero dB frequency (rad/s) of the magnitude frequency response curve for the cascaded analog compensator and plant.
- Cascade compensator design procedure:
  - Design the analog compensator $G_c(s)$ in the s-plane to meet the required performance specifications.
  - Use the Tustin transformation to obtain the model for an equivalent digital controller $G_p(z)$
  - Design verification via simulation.

Cascade compensation: example

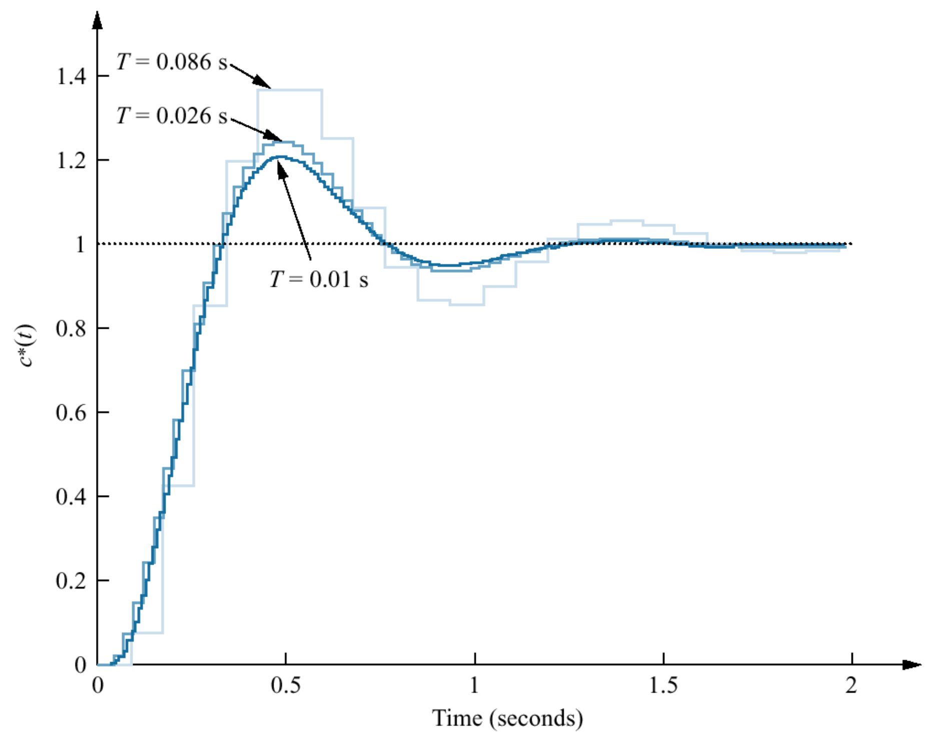
- Design a digital lead compentor for the below system, where $G_p(s) = \frac{1}{s(s+6)(s+10)}$ so that the system will operate with 20% overshoot and a settling time of 1.1 seconds.



- **Solution**:
  - Design the analog lead compensator (ELT3051): $G_c(s) = \frac{1977(s+6)}{s+29.1}$.
  - Find the zero dB frequency, ω_{Φ_M}, for $G_p(s)G_c(s)$ (ELT3051): $\omega_{\Phi_M} = 5.8$ rad/s $\rightarrow$ the lowest value of T should be in the range $\frac{0.15}{\omega_{\Phi_M}} = 0.026$ to $\frac{0.5}{\omega_{\Phi_M}} = 0.086$ seconds.

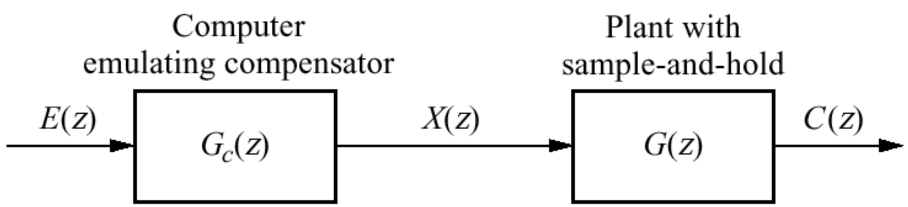
Cascade compensation: example solution

- Let choose $T = 0.01$ second, applying Tustin transformation yields $G_c(z) = \frac{1778z-1674}{z-0.746}$. The transfer function of the plant and zero-order hold is $G_p(z) = \frac{1.602z^2+1.602\times10^{-7}z+1.602\times10^{-7}}{z^3-2.847z^2+2.699z-0.8521}$.
- The simulation shows that the compensated closed-loop system meets the transient response requirements.



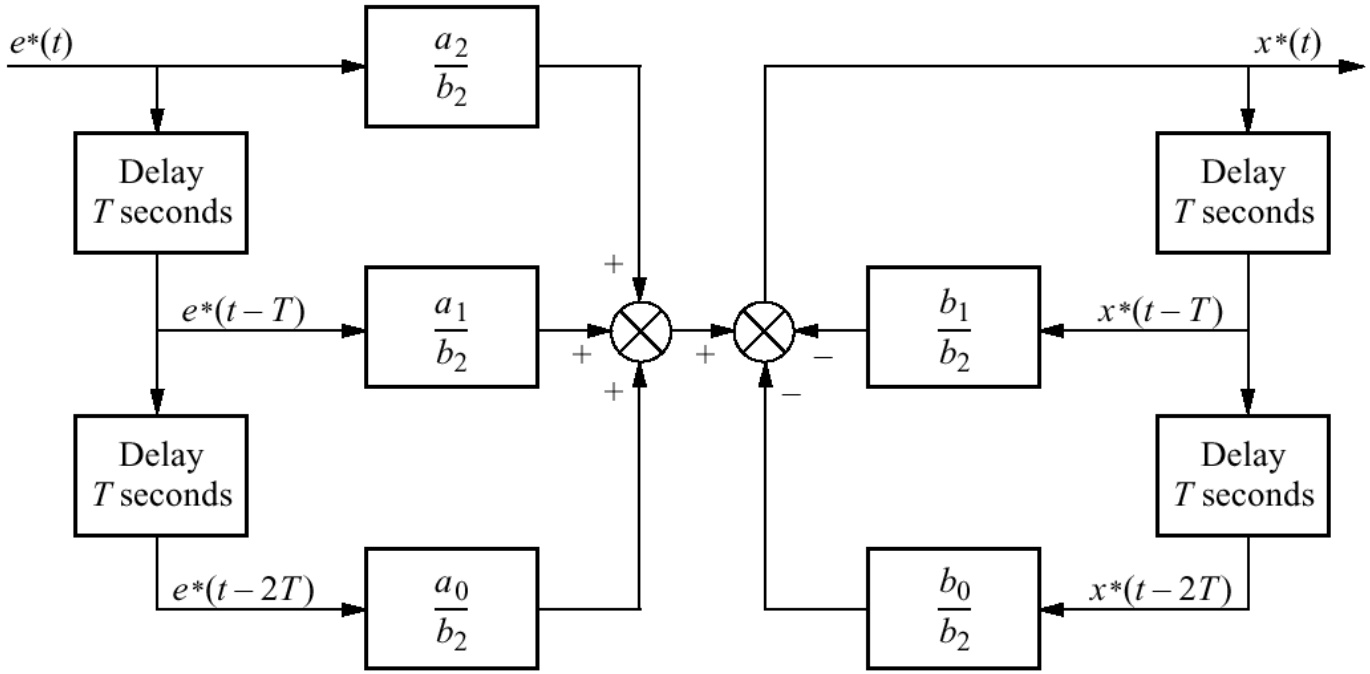
Note: Valid only at integer values of sampling instant

Implementing the digital compensator



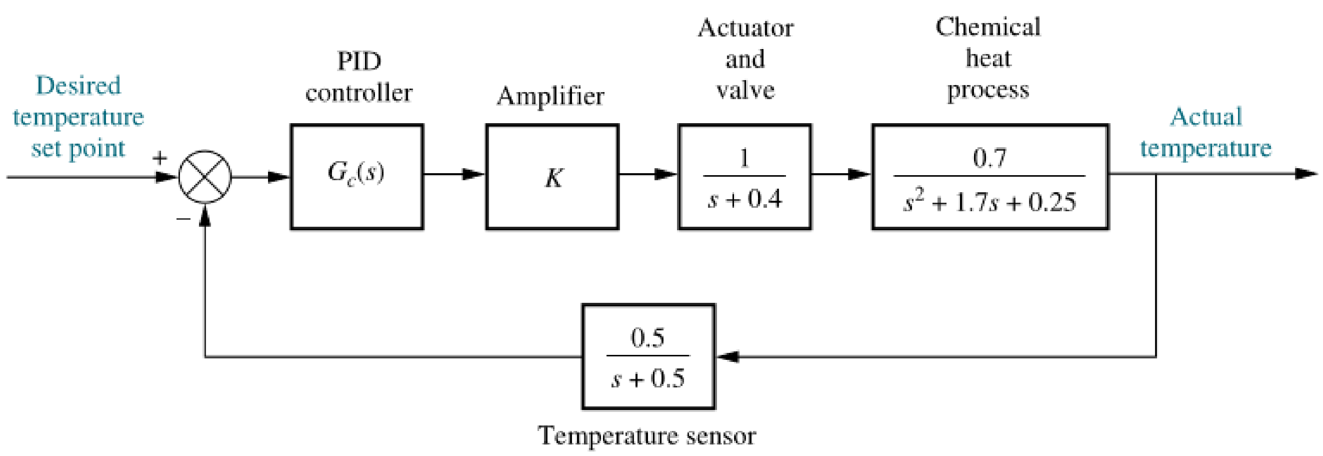
- How do we derive a numerical algorithm that the computer can use to emulate the compensator?
  - Via the computer's sampled output $x^*(t)$, whose transform is $X(z)$.
- Let $G_c(z) = \frac{X(z)}{E(z)} = \frac{a_2 z^2 + a_1 z + a_0}{b_2 z^2 + b_1 z + b_0}$
  - $X(z) = \left(\frac{a_2}{b_2} + \frac{a_1}{b_2} z^{-1} + \frac{a_0}{b_2} z^{-2}\right) E(z) - \left(\frac{b_1}{b_2} z^{-1} + \frac{b_0}{b_2} z^{-2}\right) X(z)$.
  - $x^*(t) = \frac{a_2}{b_2} e^*(t) + \frac{a_1}{b_2} e^*(t-T) + \frac{a_0}{b_2} e^*(t-2T) - \frac{b_1}{b_2} x^*(t-T) - \frac{b_0}{b_2} x^*(t-2T)$: a function of current and past samples of input & past samples of the output.
  - The compensator can be implemented by storing several successive values of the input and output; then the output is formed by a weighted linear combination of these stored variables.

Implementing the digital compensator (2)

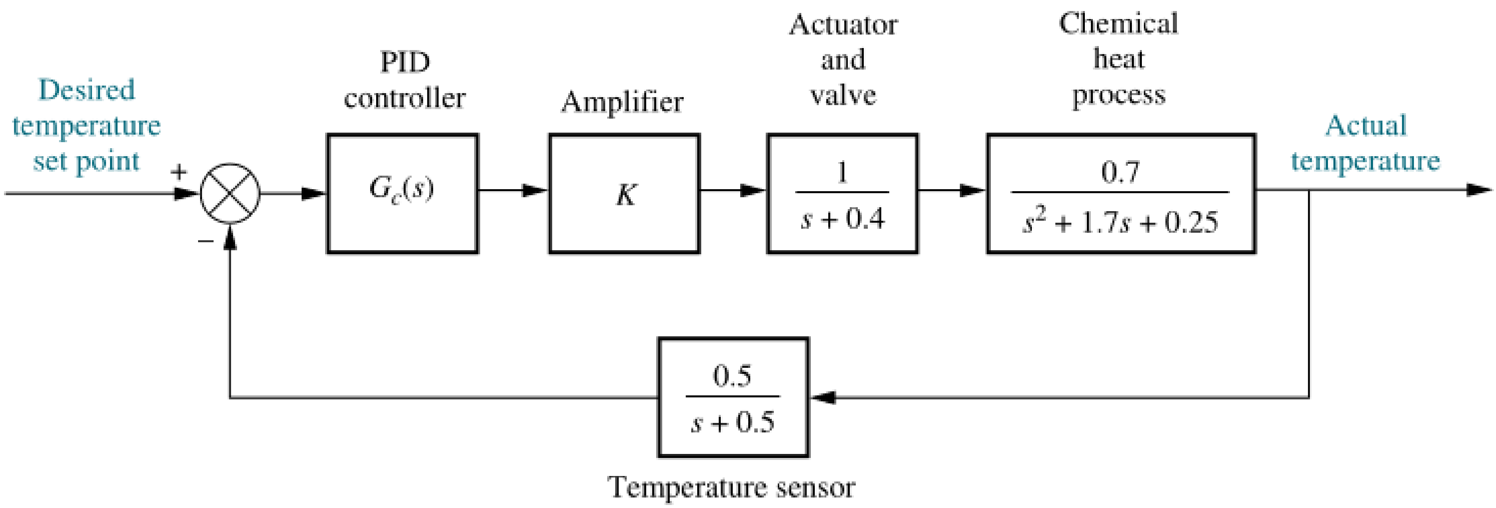


Example

- Consider the temperature control system for a chemical process shown in the below Figure. The uncompensated system is operating with a rise time approximately the same as a second-order system with a peak time of 16 seconds and 5% overshoot. Design a PID controller so that the compensated system will have a rise time approximately equivalent to a second-order system with a peak time of 8 seconds and 5% overshoot, and zero steady-state error for a step input.



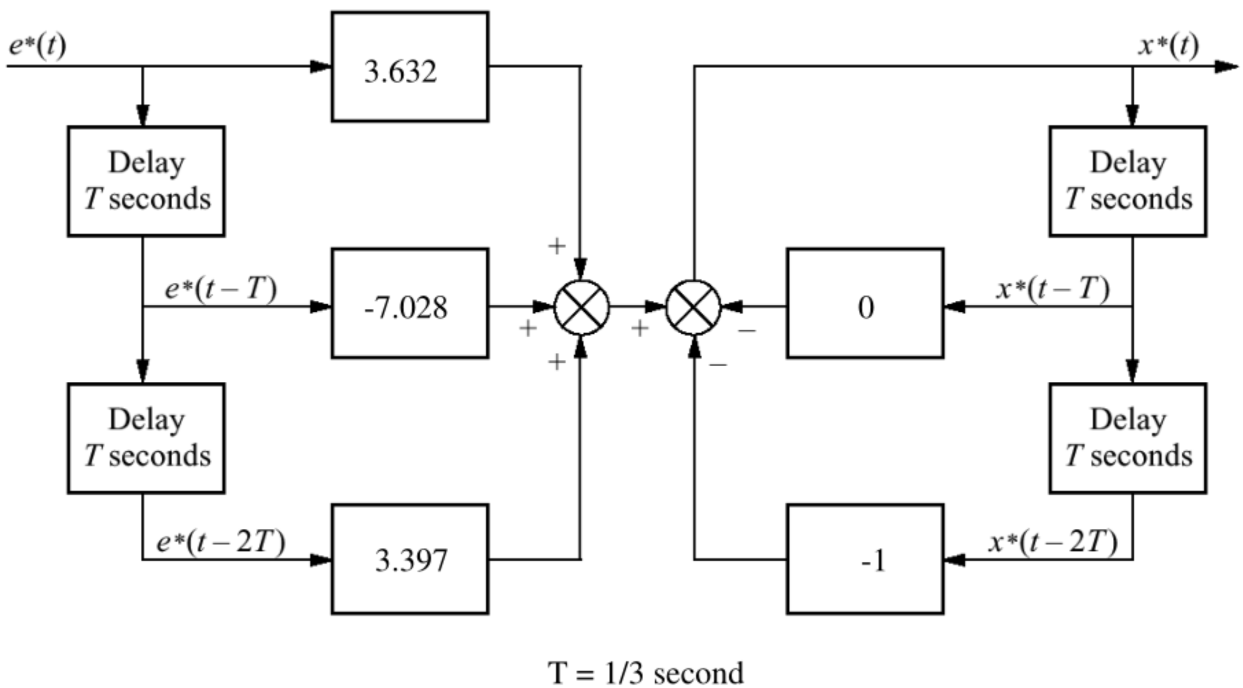
Example solution (1)



- Uncompensated: $G(s)H(s) = \frac{0.35K}{(s+0.4)(s+0.5)(s+0.163)(s+1.537)}$, searching the $133.64°$ line (%OS = 5%) & $T_p = \frac{\pi}{\omega_d} = 16$ seconds → dominant poles are $-0.187 \pm j0.196$ with gain $0.35K = 2.88 \times 10^{-2}$.
- PD compensated: design for 8 seconds peak time and 5% overshoot (ELECT3051), we obtain $z_c = 0.19$ and $0.35K = 0.205$.
- PID compensated: assume the integral controller $G_c(s) = \frac{s+0.01}{s}$, the compensator's transfer function is $G_{\text{PID}}(s) = \frac{0.35K(s+0.19)(s+0.01)}{s} = \frac{0.5857(s+0.19)(s+0.01)}{s}$

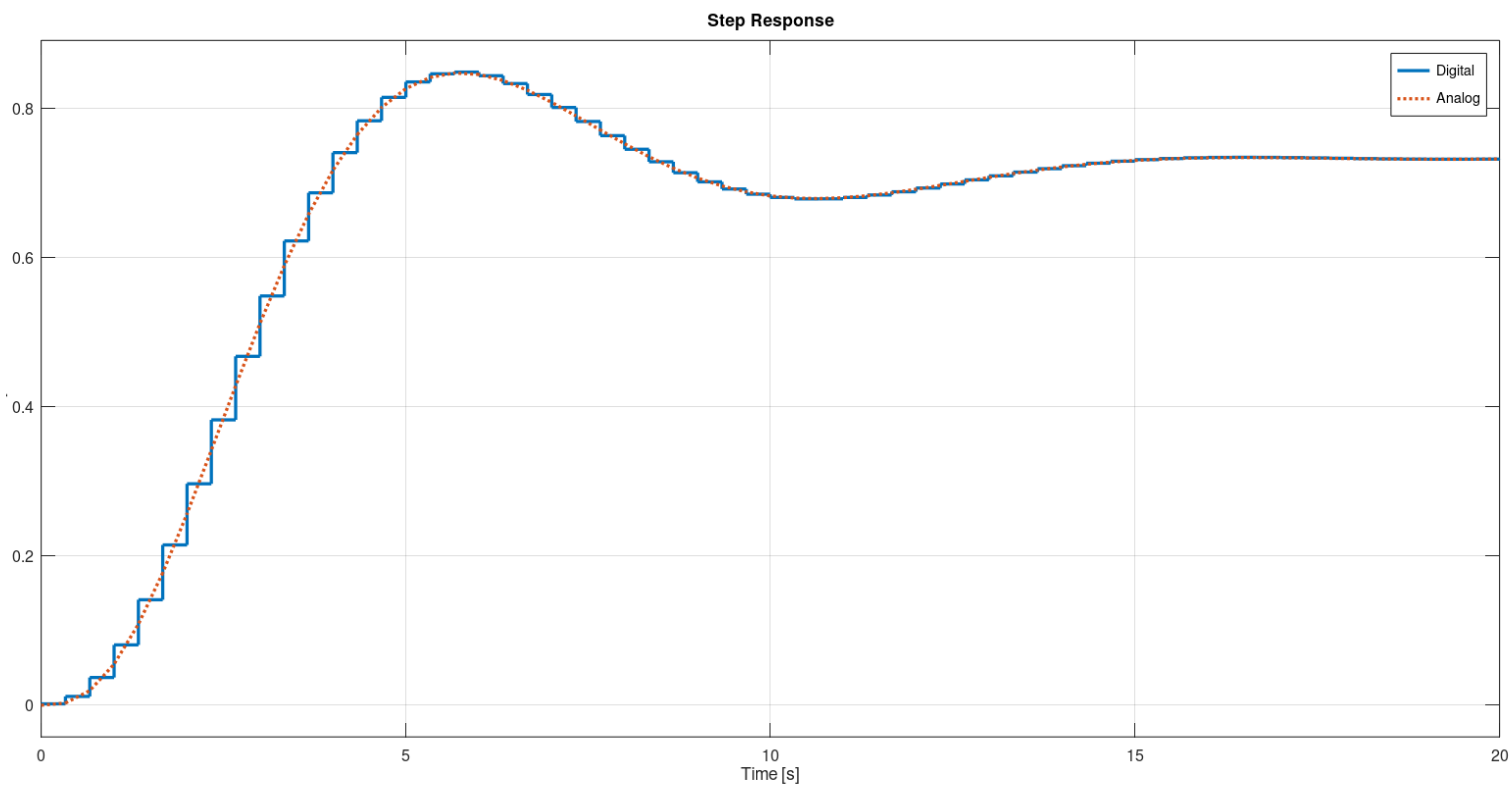
Example solution (2)

- Digitalize with $T = 1/3$ second: $G_{\mathrm{PID}}(z) = \frac{0.5857(s+0.19)(s+0.01)}{s}$
- Flowchart for implementation on a digital computer:



Example solution (3)

- Simulation verification: Homework (**will be graded**)



Summary

- Transient response analysis
  - Mapping lines on the s-plane to the z-plane through $z = e^{sT}$.
- Gain design on the z-plane
  - Stability design: search for the intersection of the root locus with the unit circle.
  - Transient response design: find the intersection of the root locus with the appropriate curves ($T\_s/T\_p/\zeta$) on the z- plane.
- Cascade compensation
  - Design the analog compensator $G_c(s)$ in the s-plane to meet the required performance specifications.
  - Use the Tustin transformation to obtain the model for an equivalent digital controller $G_p(z)$
  - Design verification via simulation.
- **Next week**: Homework (digital PID design example simulation) due.
  - Do **not** use Matlab's built-in commands for analog-digital system conversion, e.g. `c2d`, show me **your calculations** in the code.
  - Use Octave if you don not have a legitimate Matlab license!

Final project

- Apply your knowledge to implement a **simple** digital measurement & control system for real-life applications
  - Lab exercises might provide good initial suggestion (e.g. traffic light control, theft detector, motor control, etc.), but do not just copy them.
  - You can use the lab equipment for your final project; however, you don't have to stick to them, feel free to pick another platform if it suits you more.
- Assessment
  - The implemented system works as intended: 50%
  - Has all the functional blocks of a digital measurement & control system: 20%
  - Real-life potential: 10%
  - Presentation: 20%
- The goal is to learn well, getting good grades is only a symptom.
  - When you receive a low grade, worry less about how it impacts your GPA; worry more about the fact that you didn't learn as well as you should.
  - Good luck to all.